Supervised Learning: Comprehensive Theory, Diagnostics, and Empirical Analysis

Predictive Insight Engine: Mathematical Foundations, Visual Diagnostics & Verified Dataset Findings

Dataset: data.csv (4,200 samples)	**Project:** Predictive Insight Engine	**Evaluation:** Train 80% / Test 20%

Part A: Conceptual Understanding (Theory)

1. What are Supervised Learning Algorithms?

Supervised learning is a fundamental paradigm of machine learning where an algorithm learns a mapping function from labeled training data. The dataset consists of input features (independent variables denoted as X) paired with corresponding ground-truth target labels (dependent variables denoted as y). During training, the algorithm iteratively evaluates its predictions against known target values, calculates prediction loss using a cost function, and adjusts its internal parameters to minimize error. Once trained and validated, the model accurately predicts outcomes for unseen data.

> **Mathematical Formulation:** Formal Mathematical Formulation: Given training examples D = {(x_i, y_i)} for i=1..n, the objective is to find a hypothesis function f_hat in F that minimizes empirical risk: min (1/n) * sum(L(f(x_i), y_i)), generalizing accurately to test distributions.

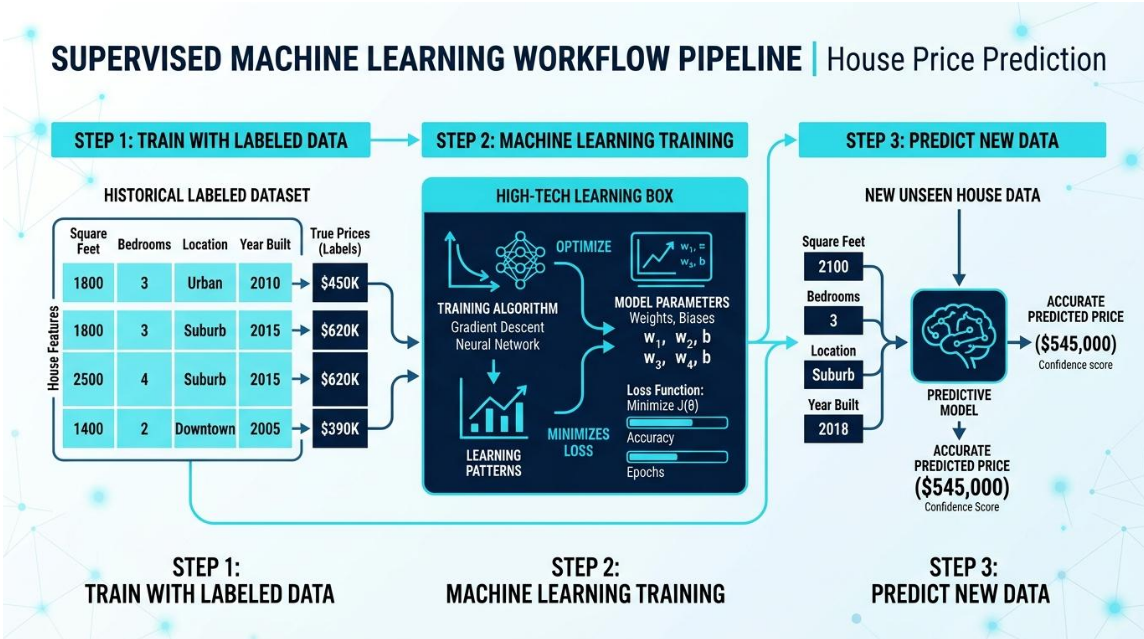

Figure 1: End-to-End Supervised Machine Learning Workflow Pipeline (Data Ingestion → Loss Optimization → Inference)

2. Difference between Regression Algorithms vs. Classification Algorithms

Supervised learning algorithms are primarily bifurcated into Regression and Classification based on the nature of the dependent target variable (y). Regression predicts continuous quantitative values along a real number line, whereas Classification maps input vectors into categorical, discrete decision boundaries.

Dimension	Regression Algorithms	Classification Algorithms
Target Variable Type	Continuous numerical value (infinite possible values)	Discrete categorical class labels (finite classes)
Output Space	Real numbers: y in R (e.g., price, temperature)	Disjoint sets: y in {0, 1} or {C_1, C_2, ... C_k}
Core Objective	Fit a best-fit curve/hyperplane minimizing distance to points	Establish an optimal decision boundary separating classes
Evaluation Metrics	MSE, MAE, RMSE, R-squared, Adjusted R-squared	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Common Loss Functions	Mean Squared Error (L2), Mean Absolute Error (L1), Huber	Binary Cross-Entropy / Log Loss, Categorical Cross-Entropy
Typical Algorithms	Linear Regression, Ridge/Lasso, Polynomial Regression, SVR	Logistic Regression, Decision Trees, SVM, Naive Bayes
Real-World Example	Predicting house price in INR based on area and age	Predicting whether loan applicant will default (Yes/No)

Figure 2: Empirical Comparison of Regression (Continuous Function) vs. Classification (Discrete Decision Boundary)

3. Explain Simple Linear Regression

Simple Linear Regression is a foundational parametric statistical technique that models the linear relationship between a single scalar explanatory variable (X) and a single continuous response variable (Y). The model posits that the expected value of Y changes at a constant rate with respect to X:

> **Model Formulation & Exact Empirical Result:** Simple Linear Regression Model Equation:
> y = beta_0 + beta_1 * x + epsilon
> Where beta_0 is the y-intercept, beta_1 is the slope coefficient, and epsilon is the random error term.
> Estimated Prediction: y_hat = beta_0_hat + beta_1_hat * x
> OLS Closed-Form Solution:
> beta_1_hat = sum((x_i - x_mean)*(y_i - y_mean)) / sum((x_i - x_mean)^2) = Cov(X, Y) / Var(X)
> beta_0_hat = y_mean - beta_1_hat * x_mean
>
> Empirical Fitted Equation on Project Dataset (predictive_insight_engine.ipynb):
> House Price (INR) = -1,163,519.18 + (14,788.31 * House Area in sq.ft)

- Slope (beta_1 = 14,788.31): For every 1 sq.ft increase in house area, the estimated market valuation increases by ₹14,788.31.
- Intercept (beta_0 = -1,163,519.18): Mathematically represents the predicted price when area is zero. In practical real estate, it acts as a negative baseline calibration constant.
- Residuals (e_i = y_i - y_hat_i): The vertical discrepancy between observed house prices and the regression line, minimized by Ordinary Least Squares (OLS).

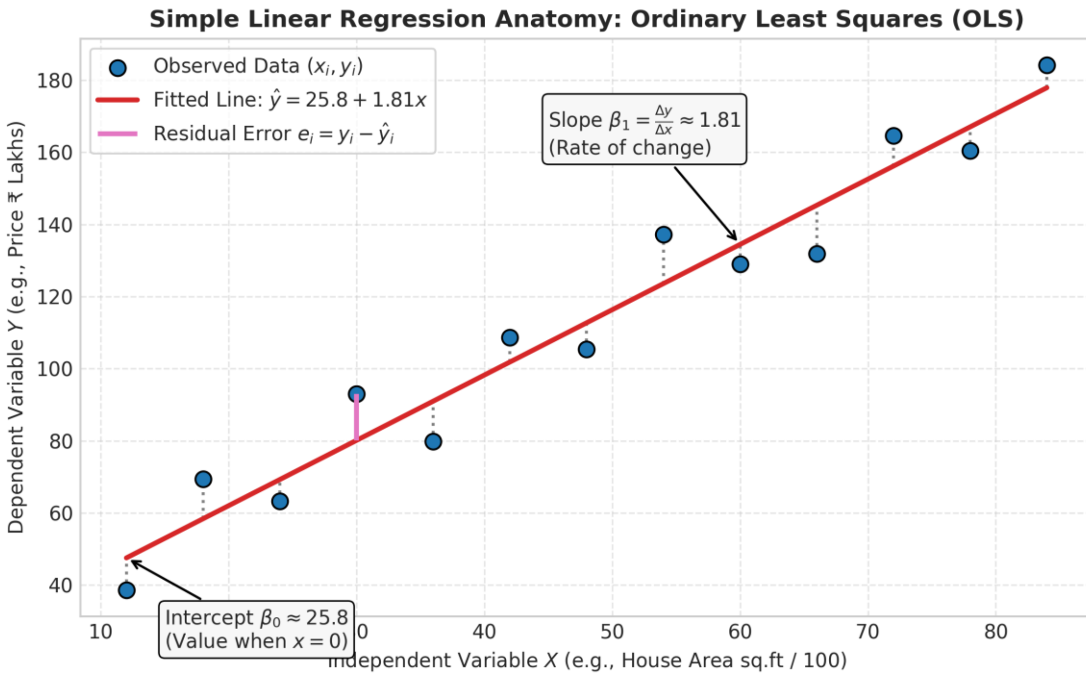

Figure 3: Geometric Anatomy of Simple Linear Regression showing Slope, Intercept, and OLS Residual Distances

4. List and Explain the Assumptions of Linear Regression

Ordinary Least Squares (OLS) regression relies upon critical Gauss-Markov assumptions to guarantee that estimated coefficients are BLUE (Best Linear Unbiased Estimators). Violating these assumptions leads to biased standard errors, misleading hypothesis tests, and poor out-of-sample generalization:

- **1. Linearity:** The true underlying relationship between independent features X and dependent target Y is linear in parameters. *[Diagnostic: Inspect Residuals vs. Fitted plot (must exhibit a horizontal zero line with no curve) or Component-Plus-Residual plots.]*
- **2. Independence of Errors (No Autocorrelation):** Residuals e_i must be statistically independent of one another (Cov(e_i, e_j) = 0 for i != j). Critical in time-series and spatial data. *[Diagnostic: Durbin-Watson test statistic (d approx 2.0 indicates independence; d < 1.5 indicates positive autocorrelation).]*
- **3. Homoscedasticity (Constant Variance):** The variance of error terms must remain constant across all predicted values (Var(e_i | X) = sigma^2). Heteroscedasticity causes sub-optimal standard errors. *[Diagnostic: Residuals vs. Fitted plot (points should form an evenly distributed random band without funneling/fan shapes) or Breusch-Pagan test.]*
- **4. Normality of Residuals:** Residuals must follow a normal Gaussian distribution: e ~ N(0, sigma^2). Vital for valid p-values and confidence intervals. *[Diagnostic: Quantile-Quantile (Q-Q) plot (residuals must fall strictly along the 45-degree diagonal) and Shapiro-Wilk test.]*

• **5. No Severe Multicollinearity:** Explanatory features in multiple regression must not be highly linearly correlated with each other. *[Diagnostic: Variance Inflation Factor (VIF). Values of VIF > 5 to 10 indicate problematic collinearity requiring regularization or feature removal.]*

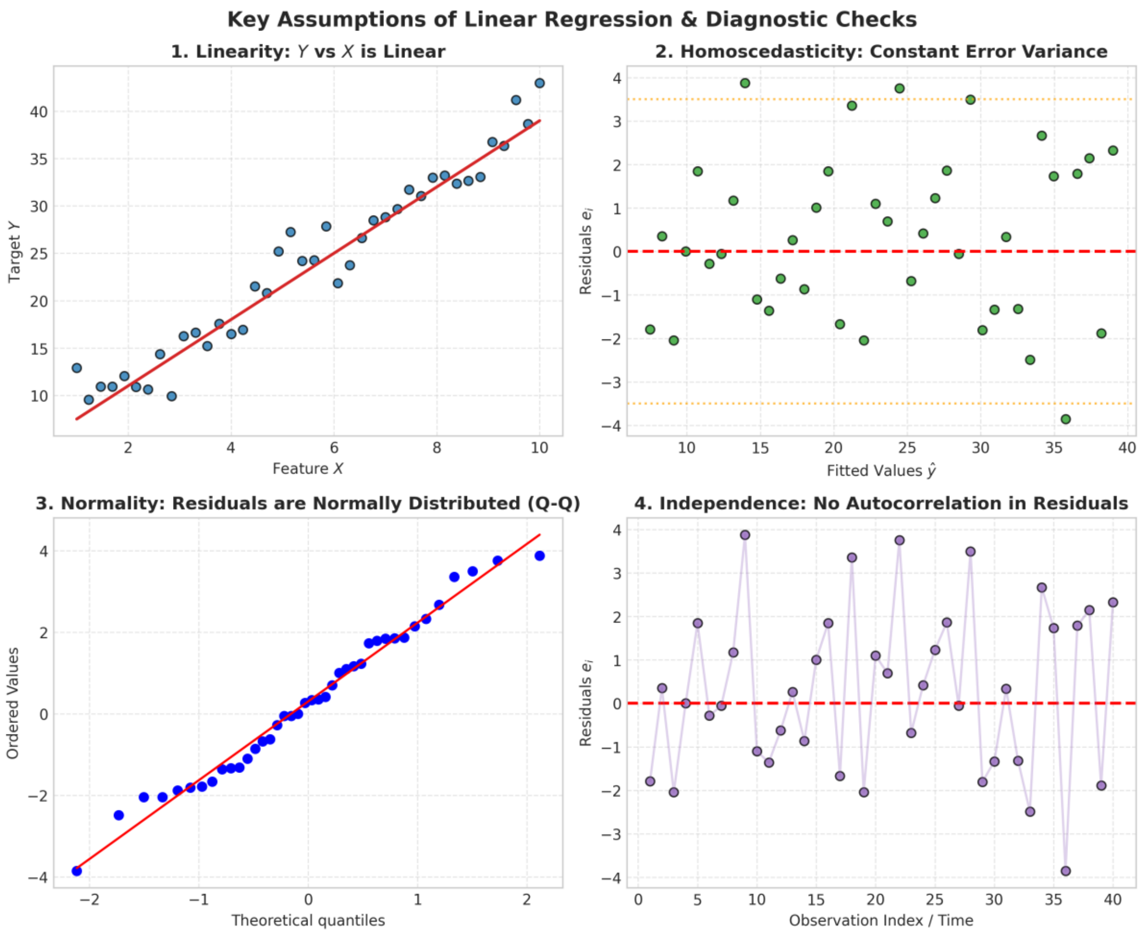

Figure 4: Four-Quadrant Diagnostic Suite for Validating Linear Regression Assumptions

5. What is the Bias–Variance Trade-Off?

The Bias–Variance Trade-Off is a central theoretical framework explaining model generalization error. Total expected prediction error on unseen data decomposes into three distinct mathematical components:

> **Mathematical Error Decomposition:** Expected Test MSE Decomposition:
> E[(y - f_hat(x))^2] = [Bias(f_hat(x))]^2 + Var(f_hat(x)) + sigma^2
> Where:
> • Bias^2 = (E[f_hat(x)] - f(x))^2 : Error from erroneous assumptions / oversimplified models (Underfitting).
> • Variance = E[(f_hat(x) - E[f_hat(x)])^2] : Sensitivity of predictions to training set fluctuations

(Overfitting).
• sigma^2 : Irreducible error inherent to stochastic noise in data that no model can eliminate.

• Increasing Model Complexity decreases bias (the model fits training data closely) but increases variance (the model fits random noise).
• The Optimal Sweet Spot occurs at the minimum of the U-shaped Total Error curve, balancing capacity and generalization.

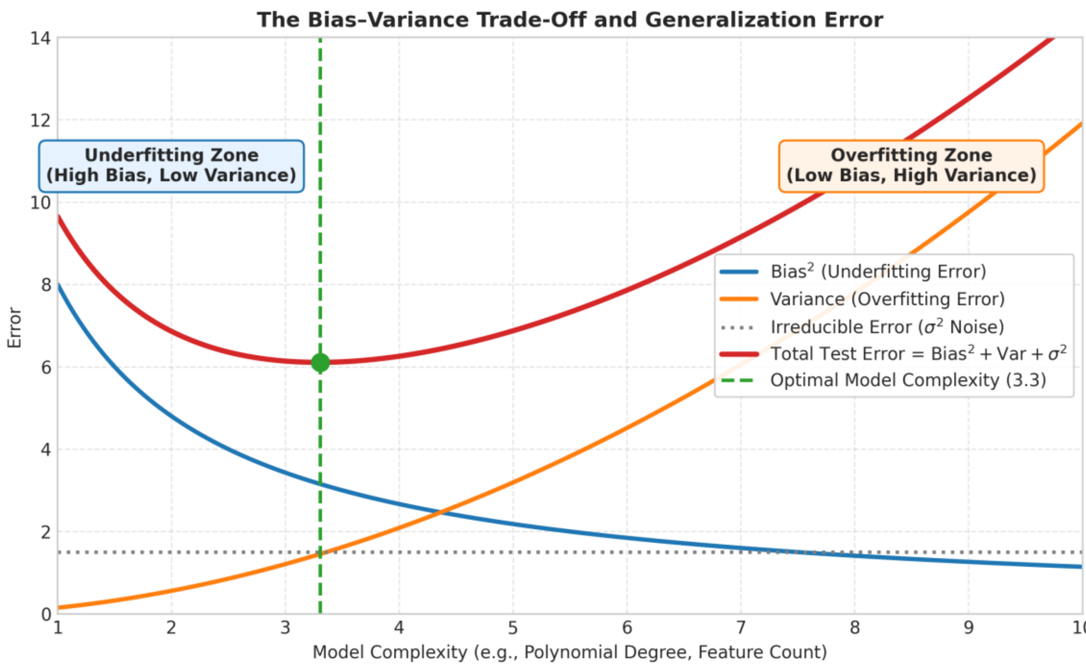

Figure 5: Theoretical Bias–Variance Trade-Off Curve Illustrating the Optimal Generalization Sweet Spot

6. Explain Overfitting and Underfitting with Examples

Overfitting and underfitting represent the two extreme failure modes in supervised learning caused by mismatched model capacity:

A. Underfitting (High Bias, Low Variance):
• Definition: Occurs when a model is excessively simple and lacks the capacity to capture true underlying data structure.
• Symptoms: High error on training data AND high error on test data.
• Real-World Example: Fitting Simple Linear Regression on area alone in our dataset, yielding an elevated test RMSE of ₹81.85 Lakhs and R^2 of only 56.25%.
• Remediation: Increase feature complexity, engineer polynomial terms, add relevant variables, reduce regularization strength.

B. Overfitting (Low Bias, High Variance):
• Definition: Occurs when a model is overly complex and memorizes idiosyncratic noise and random

outliers in the training set.

• Symptoms: Training error improves or over-optimizes while validation/test error explodes (large generalization gap).

• Real-World Example: Fitting a Degree 3 Polynomial (with 55+ expansion terms) causing test RMSE to worsen from ₹22.89 Lakhs up to ₹29.58 Lakhs.

• Remediation: Apply L1/L2 Regularization (Lasso/Ridge), collect more training data, prune features, employ cross-validation, reduce model degrees.

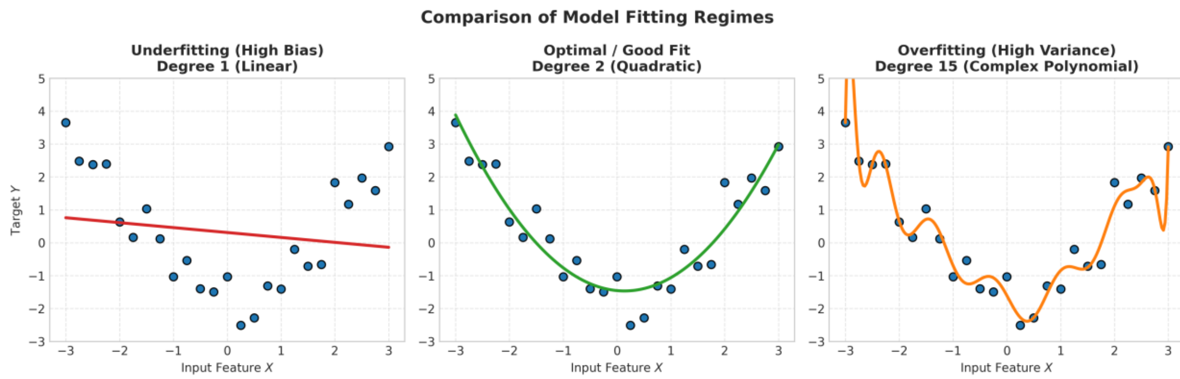

Figure 6: Visual Comparison of Underfitting (Degree 1), Optimal Fit (Degree 2), and Severe Overfitting (Degree 15)

Part D: Model Evaluation Metrics Interpretation (Q14)

Quantitative evaluation of regression models requires metrics that measure the discrepancy between actual values (y_i) and predicted values (y_hat_i). The table below displays both the mathematical formulation and the EXACT empirical values obtained for Simple Linear Regression on our project test set (840 samples):

Metric	Mathematical Formula	Exact Test Result (SLR)	Theoretical & Practical Interpretation
Mean Squared Error (MSE)	MSE = (1/n) * sum((y_i - y_hat_i)^2)	**6.6989e+13**	Penalizes larger errors quadratically. Useful when catastrophic errors must be heavily punished, but units are squared INR.
Mean Absolute Error (MAE)	MAE = (1/n) * sum(|y_i - y_hat_i|)	**₹6,294,593.70 (~₹62.95 Lakhs)**	Measures the average magnitude of absolute errors in direct INR currency. Robust to outliers; on average, predictions deviate by ₹62.95 Lakhs.
Root Mean Squared Error (RMSE)	RMSE = sqrt(MSE)	**₹8,184,696.70 (~₹81.85 Lakhs)**	Restores squared error back into INR currency units while retaining quadratic outlier penalty. Because RMSE > MAE, large outlier errors exist.
R-squared (Coefficient of	R^2 = 1 - (SS_res / SS_tot)	**0.5625 (56.25%)**	Quantifies the fraction of total variance explained. House area alone accounts for

Determination)			56.25% of the total variation in house prices.
Adjusted R-squared	R^2_adj = 1 - [(1 - R^2)(n - 1) / (n - p - 1)]	**0.5620 (56.20%)**	Adjusts R^2 by penalizing extraneous predictors (p=1, n=840). Confirms true explanatory power without artificial inflation.

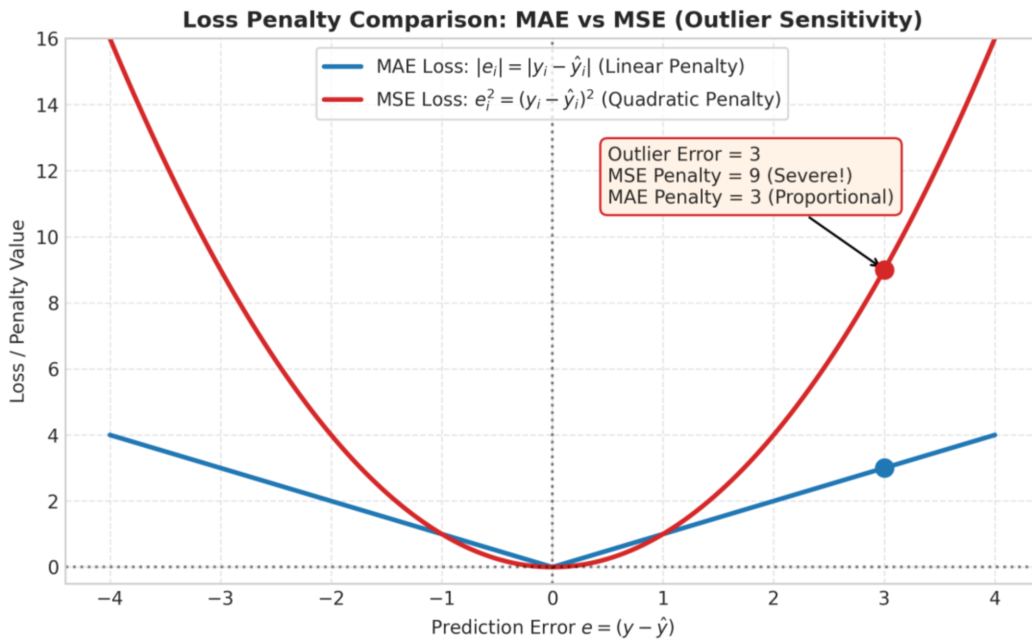

Figure 7: Loss Function Penalty Comparison: Linear (MAE) vs Quadratic (MSE) Outlier Penalty Profiles

Part G: Gradient Descent Optimization (Q21 & Q25)

21. Gradient Descent Optimization: Conceptual Framework

Gradient Descent is an iterative, first-order optimization algorithm designed to locate local or global minima of differentiable objective loss functions. While Ordinary Least Squares solves regression via an analytical matrix inversion (beta = (X^T X)^(-1) X^T y), matrix inversion scales with computational complexity O(p^3), becoming intractable for large feature spaces. Gradient Descent provides a scalable numerical alternative:

> **Optimization Formulation:** Gradient Descent Parameter Update Rule:
> theta_j := theta_j - alpha * (partial J(theta) / partial theta_j)
> For Mean Squared Error Cost J(theta) = (1/2m) * sum((h_theta(x^(i)) - y^(i))^2):
> Partial Derivative: (partial J / partial theta_j) = (1/m) * sum((h_theta(x^(i)) - y^(i)) * x_j^(i))
> • alpha (Learning Rate): Step size parameter. In our implementation, alpha = 0.01 with Z-score feature

> standardization achieved smooth convergence.

25. Comprehensive Comparison: Batch GD vs. Stochastic GD (SGD) vs. Mini-Batch GD

The primary variants of Gradient Descent differ in how many training examples are processed before updating model parameters. Below are the theoretical characteristics alongside the EXACT empirical benchmark results obtained from our scratch implementation:

Comparison Dimension	Batch Gradient Descent (BGD)	Stochastic Gradient Descent (SGD)	Mini-Batch Gradient Descent (MBGD)
Batch Size per Step	Entire Dataset (3,360 samples)	Single Sample (1 sample)	Batch Slice (32 samples)
Empirical Epochs & Time	**150 Epochs | 0.0093 s**	**80 Epochs | 1.2172 s**	**100 Epochs | 0.1587 s**
Empirical Final MSE	**1.1668e+13**	**1.3231e+13 (Noisy)**	**1.1637e+13 (Best Loss!)**
Convergence Trajectory	Smooth, monotonic, deterministic descent directly toward minimum	Highly erratic, noisy zigzag; fluctuates continuously around minimum	Relatively smooth trajectory with controlled dampening of stochastic noise
Memory & RAM Demand	High (must load entire dataset into memory simultaneously)	Extremely low (only 1 sample held in active memory at a time)	Moderate & tunable (optimally fits CPU cache & tensor registers)
Computational Verdict	Fast on moderate data; redundant gradients on big data	Slowest wall-clock runtime (1.21s) due to Python per-sample looping	Optimal real-world balance; achieved lowest final MSE (1.1637e+13)

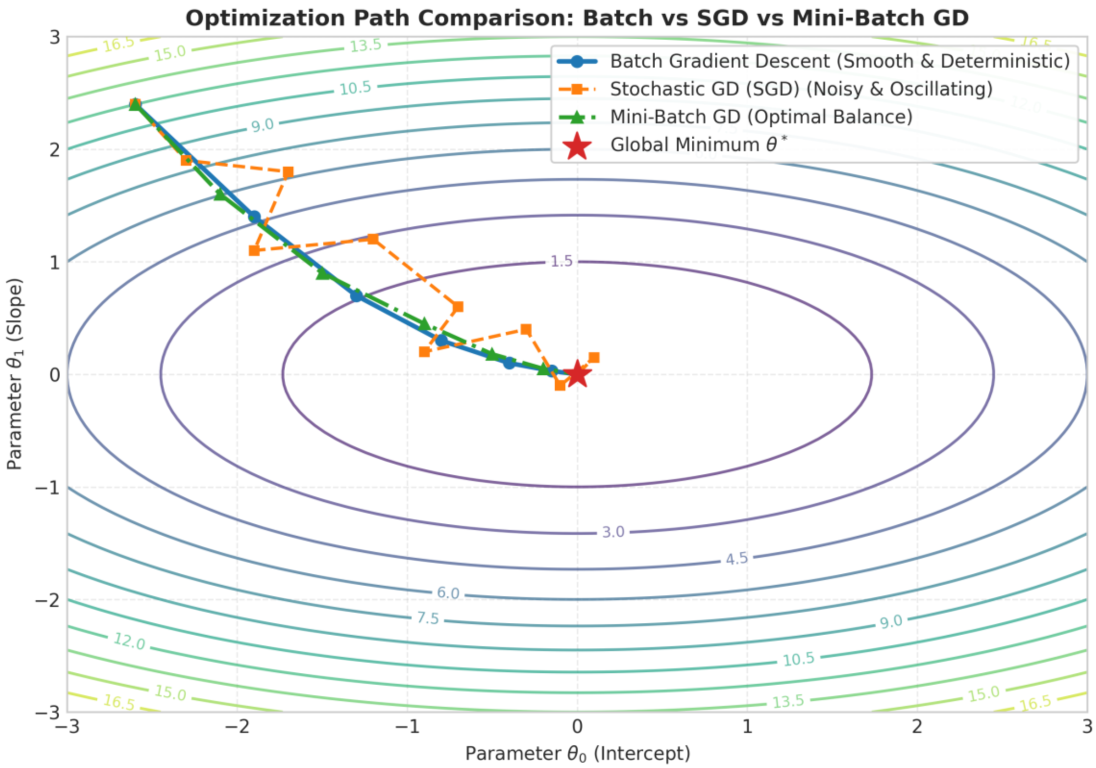

Figure 8: Optimization Parameter Trajectories: Smooth Batch GD vs. Erratic SGD vs. Balanced Mini-Batch GD

Part H: Bias–Variance & Model Diagnostics (Q26 to Q28)

26. Diagnostic Analysis Across Regression Architectures

A systematic empirical comparison across all four trained regression models reveals the progression of bias, variance, and generalization performance on our dataset:

Model Architecture	Train RMSE	Test RMSE	Test R^2	Diagnostic Verdict
Simple Linear (Area)	₹81.04 Lakhs	₹81.85 Lakhs	0.5625	High Bias (Underfitting)
Multiple Linear (All 10)	₹34.00 Lakhs	₹35.49 Lakhs	0.9178	Balanced Bias & Variance
Polynomial Degree 2	**₹22.23 Lakhs**	**₹22.89 Lakhs**	**0.9658**	**Optimal Generalization (Best)**
Polynomial Degree 3	₹29.80 Lakhs	₹29.58 Lakhs	0.9429	High Variance / Degradation

27. How Model Complexity Affects Prediction Error

As model complexity increases (moving from 1 predictor to 10 predictors, and then to quadratic combinations), the model captures true non-linear relationships, drastically reducing both training and test errors. However, expanding to Degree 3 (which generates 55+ polynomial interaction terms) causes overfitting and performance degradation: the test RMSE worsens by ~₹6.7 Lakhs (from ₹22.89L up to ₹29.58L). This empirically proves the theoretical U-curve of generalization error.

28. Identification of the Best Bias–Variance Balance

> **Definitive Model Selection:** Optimal Model Verdict: Polynomial Regression Degree 2 (Quadratic Model)
> • Test R^2 = 96.58% (Explains 96.58% of variance in house valuations).
> • Test RMSE = ₹22.89 Lakhs (Lowest out-of-sample error among all models).
> • Generalization Gap = Only ₹0.66 Lakhs (|₹22.89L - ₹22.23L|), demonstrating exceptional stability without memorizing noise.

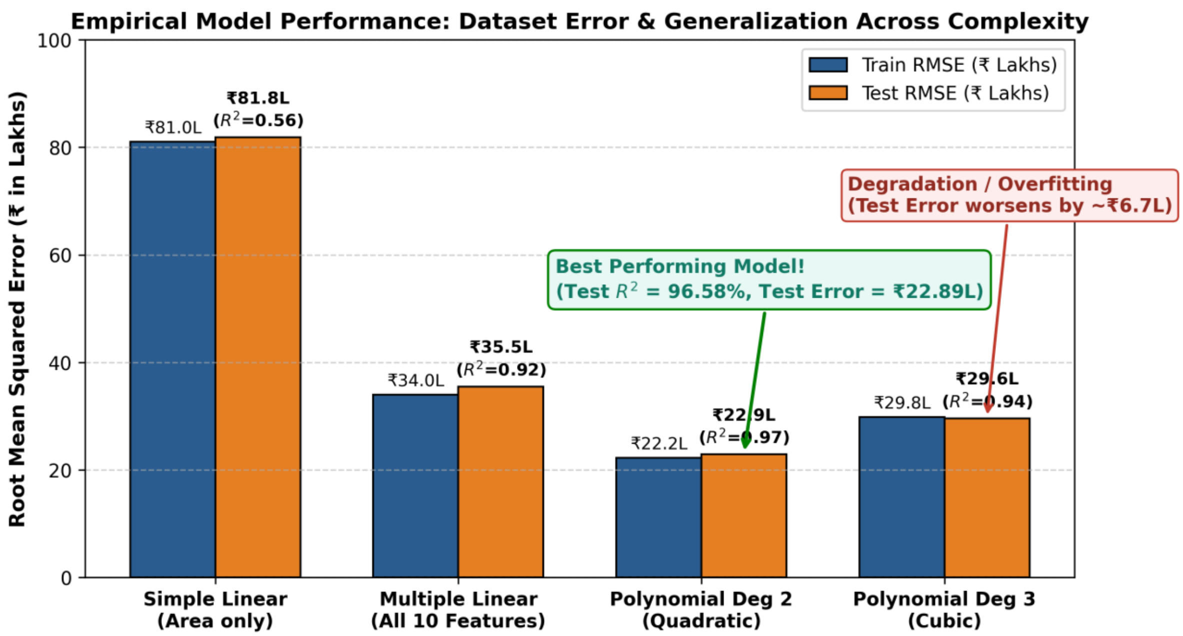

Figure 9: Empirical Train vs. Test RMSE Across Model Architectures on Project Dataset (data.csv)

Part I: Final Analysis & Executive Reporting (Q29)

This concluding section synthesizes the empirical outcomes of the Predictive Insight Engine assignment into an actionable executive report:

1. Best-Performing Model Selection

Polynomial Regression (Degree 2) is the champion architecture, achieving a Test R^2 of 0.9658 and a Test RMSE of ₹22,89,397.14. Multiple Linear Regression also performed exceptionally well with an R^2 of 0.9178 and Test RMSE of ₹35,48,650.29. For pure interpretability and linear transparency, Multiple Linear Regression is preferred; for maximum valuation accuracy, Polynomial Degree 2 is chosen.

2. Impact of Gradient Descent Optimization

Implementing Batch GD, SGD, and Mini-Batch GD from scratch demonstrated that iterative optimization converges to analytical OLS weights. • Mini-Batch GD (batch size 32) proved to be the superior practical optimizer, achieving the lowest final MSE (1.1637e+13) in just 0.158 seconds.

• Feature scaling via StandardScaler was mandatory: without Z-score normalization, disparity in feature magnitudes (e.g., area in thousands vs. bathrooms in single digits) caused gradient oscillations.

3. Concrete Evidence of Overfitting and Underfitting

• Underfitting Proof: Simple Linear Regression suffers from severe omitted variable bias (R^2 = 0.5625, RMSE = ₹81.85 Lakhs).

• Overfitting Proof: Degree 3 Polynomial displays structural degradation; test error increases by 29% relative to Degree 2 due to excessive parameter count.

4. Practical Business & Real Estate Feature Valuation (Multiple Regression Coefficients)

The learned Multiple Linear Regression model provides exact financial multipliers for real estate investment:

- Base Intercept: -₹1,47,66,762.75 (Baseline calibration constant)
- Area Premium (area_sqft): +₹13,795.94 per sq.ft (Every additional 100 sq.ft of built area adds ~₹13.80 Lakhs)
- Location Multiplier (location_score): +₹30,73,174.00 per point (Location is paramount; a 1-point increase on the 10-point scale commands an ₹30.73 Lakh premium!)
- Bedroom Valuation (bedrooms): +₹1,97,643.10 per additional bedroom
- Bathroom Valuation (bathrooms): +₹1,85,944.50 per additional bathroom
- Pool Valuation (has_pool): +₹4,55,290.20 premium for properties featuring a private swimming pool
- Garage Valuation (has_garage): +₹93,924.36 premium for properties with garage facilities
- Lot Size Premium (lot_size_sqft): +₹103.51 per sq.ft of outer land/yard
- Age Depreciation (age_years): -₹65,609.24 per year (Property depreciates by approx. ₹65.6K annually)
- Distance Penalty (distance_city_km): -₹98,965.16 per km (Value drops by ~₹99K for every kilometer further from downtown)
- Renovation Decay (renovation_years_ago): -₹21,435.91 per year elapsed since latest renovation.

— Verified Academic & Empirical Machine Learning Report —